

Sanjin Hosic

PH.D., CHEMICAL ENGINEERING · M.S., DATA SCIENCE (IN PROGRESS)

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Summary

Founded the data science function for Novartis's Cell Therapy Technical R&D organization, transforming decision-making across 4 global sites and 10+ clinical indications. Enabled program go/no-go decisions, manufacturing optimization for commercial viability, and regulatory data packages from pre-IND through Phase 2.

Ph.D. in Chemical Engineering with a career spanning bench-scale CAR-T process development through enterprise analytics, translating statistical models into manufacturing strategy and drug development decisions.

Technical Skills

- **Analytics:** Machine Learning, Bayesian Inference, Mixed-Effects Models, NLP, Predictive Modeling, Patient Stratification, Design of Experiments (DoE)
- **Data & Platforms:** R (Advanced), Python, SQL, Shiny/Posit Connect, Git/GitLab CI/CD, Docker, Kubernetes/OpenShift, Databricks
- **High-Dimensional Data:** Flow Cytometry, Immunophenotyping, Clinical Laboratory Biomarkers, Multimodal Data Integration (Clinical, Biomarker, Manufacturing), Interactive Dashboards, ETL Pipeline Design
- **Domain:** Cell & Gene Therapy (CAR-T), Drug Development (Pre-IND through Commercial), Process Characterization, Regulatory Submissions (IND/BLA), GxP Compliance, Cross-Functional Governance

Experience

Novartis

Cambridge, MA

SENIOR EXPERT / PRINCIPAL DATA SCIENTIST – CMC

Nov 2021 - Present

- Transformed decision-making from week-long manual processes to same-day insights for cross-functional leadership by establishing the enterprise data infrastructure for Cell Therapy Technical R&D.
- Reduced manufacturing failure rates and expanded patient access by applying machine learning to identify patient-level drivers of batch failure across clinical biomarkers and manufacturing data, delivering recommendations that informed cell collection practices, process changes, and pre-treatment protocols.
- Elevated data literacy and aligned a team spanning clinical, process development, and manufacturing on analytical priorities by authoring a Statistical Analysis Plan.
- Transformed decision-making across 4 global sites and 10+ clinical indications, reducing recurring reporting from days to under 1 hour, by replacing manual CSV-based workflows with live interactive dashboards.
- Scaled data science capacity without additional FTE by deploying an internal R package and automated pipelines on OpenShift/Kubernetes, replacing ad-hoc analyst workflows with governed, reproducible infrastructure.
- Unified clinical, manufacturing, and operations data for the first time by leading the data science workstream for an executive-sponsored initiative, building the integration layer, data dictionaries, and automated cloud pipelines.
- Informed process selection and release specifications for IND filings by switching from frequentist to Bayesian methods to quantify the probability of future manufacturing success.

bluebird bio

Cambridge, MA

SCIENTIST II, CELL THERAPY PROCESS DEVELOPMENT

Jan 2020 - Nov 2021

Promoted from Scientist (Jan 2020 - Jan 2021)

- Developed closed and automated CAR-T manufacturing processes across 3 programs: a pre-clinical allogeneic gamma delta T-cell therapy, a dual-CAR program (bbT369), and the commercially approved anti-BCMA therapy ABECMA (ide-cel).
- Designed and executed DoE studies as an individual contributor, generating analytical data via multiplexed flow cytometry (Miltenyi MACSQuant) and analyzing results via multiple linear regression to drive iterative process optimization.
- Managed a team of 3 direct reports overseeing company-wide cell material supply and led cross-functional process development in a matrixed GxP environment.

Sarepta Therapeutics

Greater Boston Area

SCIENTIST, MANUFACTURING SCIENCE & TECHNOLOGY

May 2019 - Jan 2020

- Supported commercial GMP manufacturing of EXONDYS 51, including manufacturing deviation investigations across solid-state synthesis, chromatography purification, and tangential flow filtration.

- Replaced a manual review process requiring 2–4 FTEs and accelerated root cause investigation of a recurring manufacturing deviation by building an automated pipeline to classify HPLC chromatogram peaks into product vs. impurity using machine learning.

Northeastern University

Boston, MA

GRADUATE RESEARCH ASSISTANT

Sep 2014 - Apr 2019

- Conducted translational research in biomedical microfluidics and organ-on-chip systems, applying DoE methodologies and statistical analysis. Published 6 peer-reviewed papers (3 first-author) and co-invented a patented device assembly method.

Cubist Pharmaceuticals

Lexington, MA

PROCESS ENGINEER, PROCESS DEVELOPMENT

Jul 2013 - Aug 2014

- Developed purification processes for Ceftolozane/tazobactam (Zerbaxa), a commercially approved antibiotic, evaluating trade-offs between throughput, yield, and quality metrics.

Nantero

Woburn, MA

PROCESS ENGINEER, PROCESS DEVELOPMENT

May 2011 - Jul 2013

- Optimized manufacturing processes for yield, purity, and cost, developing and scaling unit operations (stirred tank reactors, TFF, chromatography) directly applicable to biopharmaceutical manufacturing.

Education

The University of Texas at Austin

Austin, TX

MASTER OF SCIENCE, DATA SCIENCE

Expected 2027

Northeastern University

Boston, MA

DOCTOR OF PHILOSOPHY, CHEMICAL ENGINEERING

Northeastern University

Boston, MA

BACHELOR OF SCIENCE, CHEMICAL ENGINEERING

Patents & Publications

- **S. Hosic**, R.A. Koppes, S.K. Murthy, A.N. Koppes, J.R. Soucy. "Fluidic device and method of assembling same." *U.S. Patent 11,351,538* (Issued 2022).
- **S. Hosic**, A.J. Bindas, M.L. Puzan, et al. "Rapid prototyping of multilayer microphysiological systems." *ACS Biomaterials Science & Engineering* (2021).
- **S. Hosic**, W. Lake, E. Stas, et al. "Cholinergic activation of primary human derived intestinal epithelium does not ameliorate TNF- α induced injury." *Cellular and Molecular Bioengineering* (2020).
- J.R. Soucy, A.J. Bindas, R. Brady... **S. Hosic**, et al. "Reconfigurable microphysiological systems for modeling innervation and multi-tissue interactions." *Advanced Biosystems* (2020).
- J. Snyder, C.M. Wang... **S. Hosic**, et al. "Materials and microenvironments for engineering the intestinal epithelium." *Annals of Biomedical Engineering* (2020).
- M. Puzan, **S. Hosic**, C. Ghio, A. Koppes. "Enteric nervous system regulation of intestinal stem cell differentiation and epithelial monolayer function." *Scientific Reports* (2018).
- **S. Hosic**, S.K. Murthy, A.N. Koppes. "Microfluidic sample preparation for single cell analysis." *Analytical Chemistry* (2016).